

Trader Behavior Analysis Under Market Sentiment Regimes

Junior Data Scientist Assignment – Web3 Trading Team

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1. Introduction

Objective

The objective of this analysis is to study how trader behavior and performance vary under different Bitcoin market sentiment conditions (Fear, Extreme Fear, Greed, Extreme Greed). By combining historical trade-level data with the Fear & Greed Index, this study aims to uncover behavioral patterns that can inform smarter, sentiment-aware trading strategies.

Provided Datasets

- Hyperliquid Historical Trader Data
- Bitcoin Fear & Greed Index

2. Data Preprocessing

Trader Data

- Converted timestamps to datetime (IST)
- Filtered relevant trade information
- New created features:
 - **is_profitable** (A boolean (True/False) flag that tells whether a trade made money.)
 - **abs_size_usd** (The absolute value of trade size in USD.)
 - **net_pnl** (The real profit after fees.)

Sentiment Data

- Parsed date column
- Normalized sentiment classifications
- Used daily sentiment as market proxy

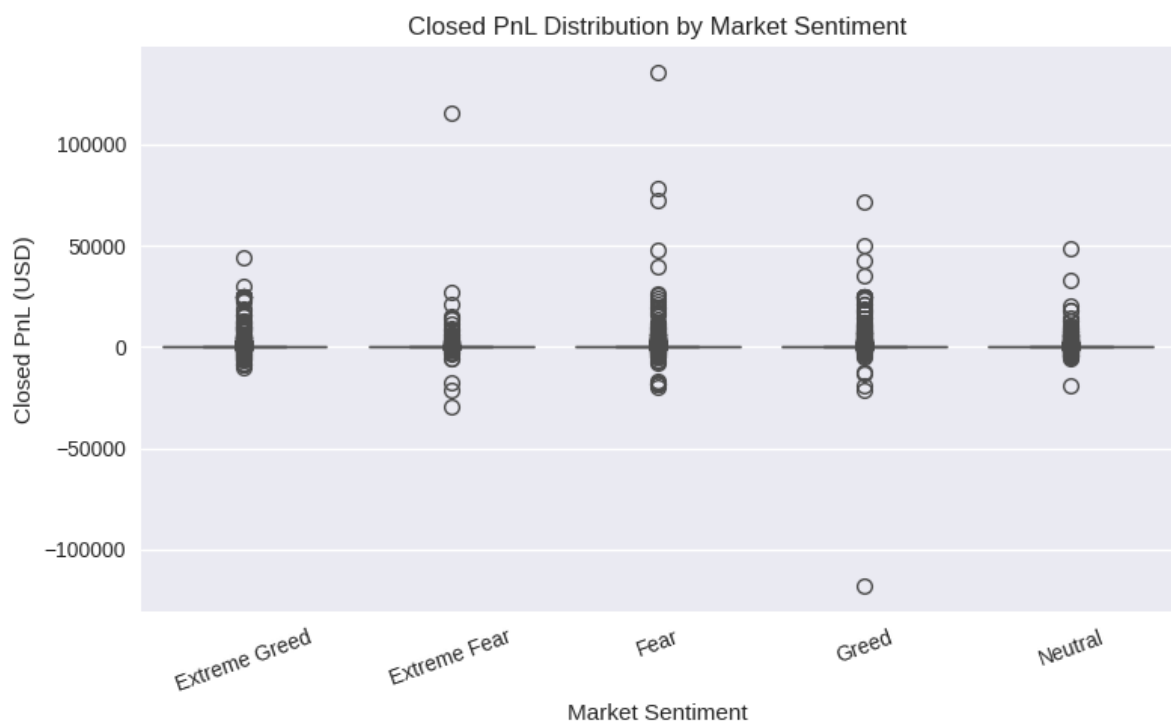
Data Merge

- Trades were merged with sentiment data using trade date
- Inner join ensured only sentiment-aligned trades were analyzed

3. Profitability vs Market Sentiment

Observation

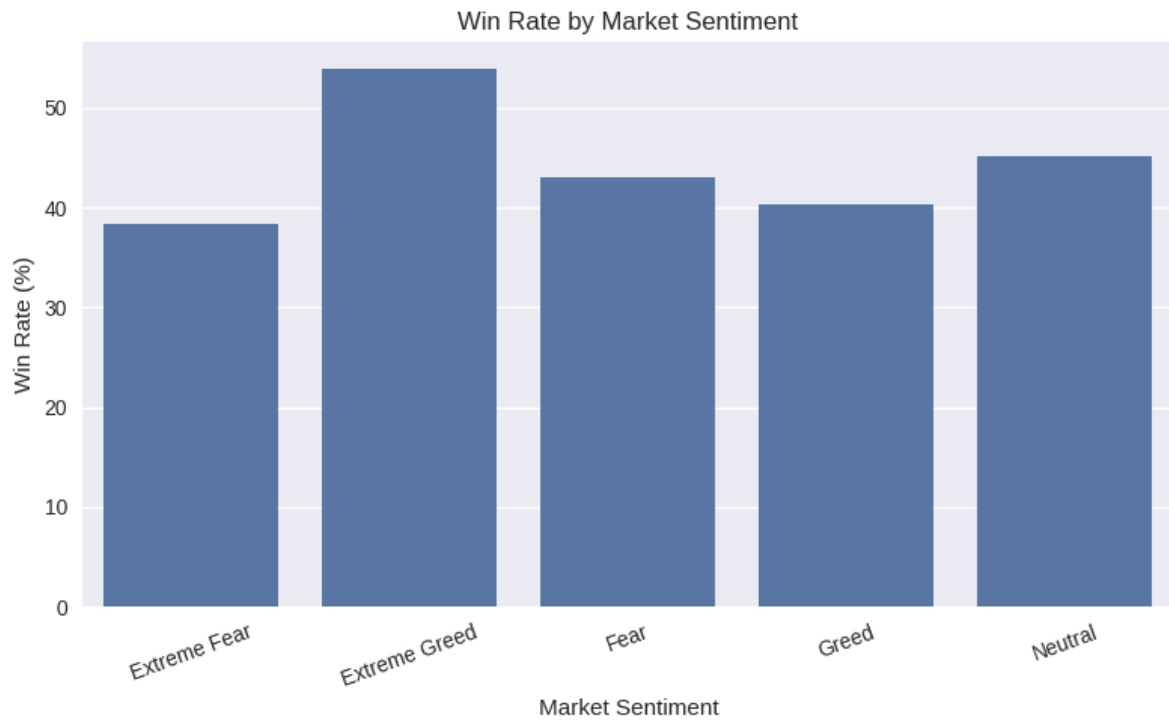
- Median and average PnL varies significantly across sentiment regimes
- Fear phases show more stable outcomes compared to Greed



Insight

Traders tend to perform more consistently during Fear periods, suggesting reduced emotional bias and better risk control.

4. Win Rate Analysis



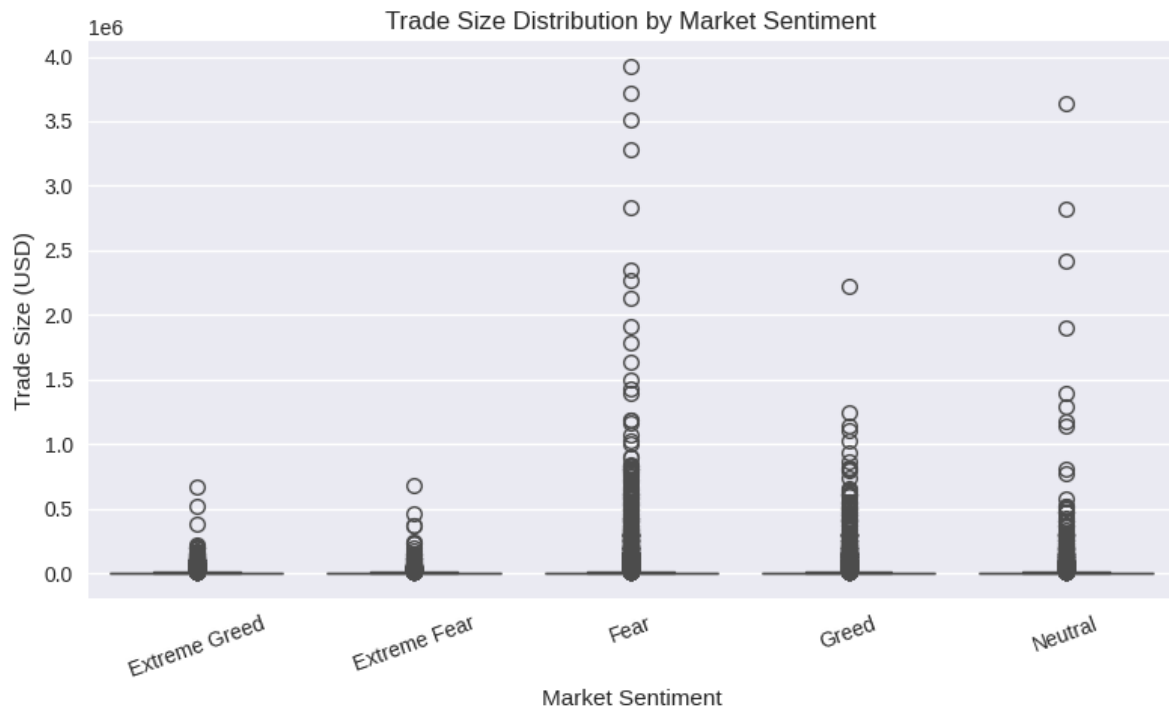
Key Finding

- Win rates are higher during Fear and Extreme Fear
- Greed phases show lower consistency

Interpretation

Emotional trading during Greed leads to overconfidence and reduced trade quality.

5. Trade Volume & Position Sizing



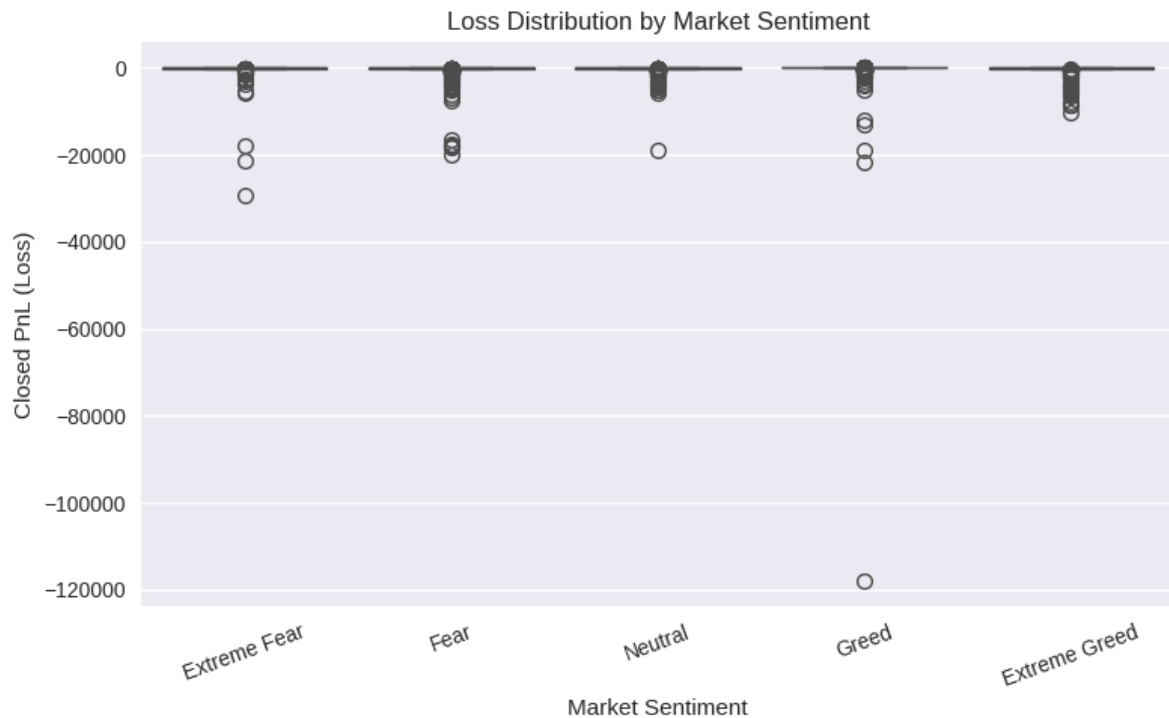
Findings

- Larger trade sizes during Greed
- More dispersed trade sizing → higher volatility

Interpretation

Traders increase exposure during optimistic market phases, increasing downside risk.

6. Loss Analysis



Key Insight

Loss magnitude is significantly higher during Greed, reinforcing the impact of overleveraging and emotional trading.

7. Business Implications

- **Sentiment-aware risk management** can reduce drawdowns
- Position sizing should be reduced during Greed phases
- Contrarian strategies may outperform during Fear periods

8. Limitations

- Analysis limited to available trade history
- Does not include liquidation events
- Sentiment is daily, not intraday

9. Conclusion

Market sentiment plays a critical role in trader behavior. This analysis demonstrates that Fear-driven markets often lead to more disciplined trading, while Greed-driven markets increase risk exposure and losses. Integrating sentiment signals into trading systems can improve risk-adjusted returns.